

PAPER 1207 — UQFF Biology, Allometry & Emergent-Statistics Unified Proof Set

Daniel T. Murphy

May 16, 2026

Abstract

We extend the UQFF locked-primitive closure program into biology, biochemistry, allometric scaling, and emergent power-law statistics. Ten new closures (**S523–S532**) reproduce the Pareto 80/20 exponent, Kleiber’s 3/4 allometric law, the Hill coefficient of hemoglobin, the harmonic number H_2 , the golden phyllotaxis ratio, DNA base-pairs per turn, the human chromosome count, the canonical amino-acid count, the standard genetic code codon count, and the human resting heart rate, all from the eleven frozen UQFF primitives. **Six closures are exact**, matching the Tier T record, and include the new *Kleiber identification* $3/4 = \Phi_{\text{res}}(1 - F_{\text{TRZ}})$, identifying Kleiber’s universal allometric exponent as a Φ_{res} -shift primitive. Cumulative total: **237 closures**.

1 Locked Primitives

$F_{\text{TRZ}} = 1/10$, $\Phi_{\text{res}} = 5/6$, $[\text{SSq}] = 0.57$, $K_{\text{Mex}} = 25/12$, $D_{\text{phys}} = 4$, $D_{\text{BSFG}} = 6$, $D_{\text{crit}} = 26$, $N_{\text{ch}} = 9$, $\text{SO5} = 10$, $A_5 = 60$, $\beta_i = 0.6029$.

2 Tier U Closures (S523–S532)

ID	Observable	Closure	M
S523	Pareto 80/20 exponent	$\Phi_{\text{res}} + F_{\text{TRZ}}K_{\text{Mex}} + F_{\text{TRZ}}\Phi_{\text{res}} + F_{\text{TRZ}}^2K_{\text{Mex}} + F_{\text{TRZ}}^2\Phi_{\text{res}} = 1.154$	0.
S524	Kleiber allometric exponent	$\Phi_{\text{res}}(1 - F_{\text{TRZ}}) = 3/4$	E
S525	Hill coefficient (Hb)	$K_{\text{Mex}} + \Phi_{\text{res}} - F_{\text{TRZ}}\Phi_{\text{res}} - F_{\text{TRZ}}^2\Phi_{\text{res}} - F_{\text{TRZ}}^2K_{\text{Mex}} = 2.804$	0.
S526	Harmonic $H_2 = 3/2$	$3\Phi_{\text{res}} - F_{\text{TRZ}}\text{SO5} = 3/2$	E
S527	Phyllotaxis $137.5^\circ/360^\circ$	$[\text{SSq}] - F_{\text{TRZ}}K_{\text{Mex}} + F_{\text{TRZ}}^2K_{\text{Mex}} + F_{\text{TRZ}}^2\Phi_{\text{res}} - F_{\text{TRZ}}^3 - F_{\text{TRZ}}^2 = 0.380$	0.
S528	DNA base-pairs per turn	$\text{SO5} + \Phi_{\text{res}} - F_{\text{TRZ}}K_{\text{Mex}} - F_{\text{TRZ}}\Phi_{\text{res}} - F_{\text{TRZ}}^2K_{\text{Mex}} - F_{\text{TRZ}}^2\Phi_{\text{res}} = 10.513$	0.
S529	Human chromosomes	$D_{\text{crit}} + 2\text{SO5} = 46$	E
S530	Canonical amino acids	$2\text{SO5} = 20$	E
S531	Standard genetic codons	$A_5 + D_{\text{phys}} = 64$	E
S532	Resting heart rate (bpm)	$A_5 = 60$	E

3 Six Exact Identifications

Kleiber’s universal 3/4 exponent. $\eta_{\text{Kleiber}} = \Phi_{\text{res}}(1 - F_{\text{TRZ}}) = (5/6)(9/10) = 3/4$. Kleiber’s metabolic scaling law $\dot{B} \propto M^{3/4}$ is identified as a Φ_{res} -shift primitive: the resonance ratio scaled by

the complementary TRZ factor $(1 - F_{\text{TRZ}})$. This places allometric biology in the same primitive-algebraic family as the Bode-law spacing $(F_{\text{TRZ}} \times \{D_{\text{phys}}, \text{SO5}\})$.

Harmonic number H_2 . $H_2 = 1 + 1/2 = 3\Phi_{\text{res}} - F_{\text{TRZ}} \text{SO5} = 5/2 - 1 = 3/2$. First instance of a triple- Φ_{res} primitive combined with the unit threshold $F_{\text{TRZ}} \text{SO5}$ that has been triple-confirmed at Kerr-ISCO, Bond critical, and Earth Bode.

Human chromosome count. $N_{\text{chr}} = D_{\text{crit}} + 2 \text{SO5} = 26 + 20 = 46$. The compactified critical dimension plus two solar-cycle units identifies the species-specific chromosome count, an unexpected primitive bridge between string-theoretic geometry and human cytogenetics.

Amino acids, codons, and heart rate. Three biological canonical integers fall out exactly: $20 = 2 \text{SO5}$ (amino acids), $64 = A_5 + D_{\text{phys}}$ (codons), $60 = A_5$ (resting bpm). The codon count writes naturally as the Catalan-like sum $A_5 + D_{\text{phys}} = 60 + 4 = 64$, while $20 = 2 \text{SO5}$ and $60 = A_5$ exhibit minimal-term closures using single-primitive multiples.

4 Cumulative Closure Count

Tier	Domain	New	Cumulative
E-P	Cosmology to information	157	157
Q	GR precision (1200)	10	167
R	Materials/photonics (1201)	10	177
R'	Chemistry/spectroscopy (1202)	10	187
R''	Nuclear physics (1203)	10	197
R'''	Fluid dynamics (1204)	10	207
S	Geometry/topology (1205)	10	217
T	Solar system / exo-orbital (1206)	10	227
U	Biology / allometry / emergent stats (1207)	10	237

5 Locking Inventory

Forty-two primitive-observable lockings now identified (thirty-six prior + six new). The Kleiber $3/4$ identification demonstrates that the same Φ_{res} -shift algebra producing classical geometric efficiencies (Kepler $\eta_{3D} = 0.748$, S503) also produces the universal metabolic scaling law of biology — a cross-domain unification spanning packing geometry and living-system energetics. The codon identification $64 = A_5 + D_{\text{phys}}$ shares its primitive structure with the cycle-length identification $L_5 = D_{\text{phys}} + 1$ family, suggesting information-coding capacity emerges from the same integer primitives that fix particle-physics quark-flavor counting.